ALL APPROACHES COMBINED vs BASELINE

Date: 2026-11-11, 15:50–17:00 UTC. Board: VisionFive 2 (JH7110, 4x SiFive U74), OpenSSL 3.5.9. Combines the three approaches tested separately (see their own folders) and compares them with the **BASELINE** in the same session.

The two configurations

	BASE (= baseline setup)	COMBINED (all three approaches)
CPU clock	default: `ondemand`, min 312.5 MHz	**approach 1:** `ondemand`, **min 750 MHz**
OpenSSL build	plain `rv64gc` (std for legacy, opt for hybrid)	**approach 2:** `-march=rv64gc_zba_zbb` (bitmanip)
Hybrid authentication (own server)	v1: ML-DSA-44 after the handshake, one core	**approach 3:** v2: ML-DSA-44 in the TLS Certificate, verified on a **second core** in parallel with ECDSA
Legacy	X25519 + RSA-3072	X25519 + RSA-3072 (same clock and build as COMBINED)

Against the hackathon server only approaches 1 and 2 apply: FOCAL's server supports the hybrid key exchange but authenticates with an RSA-3072 certificate only (ECDSA and ML-DSA signatures are refused), so the hybrid authentication (and with it approach 3) can't be used there. Hybrid there means X25519MLKEM768 key exchange.

Method

- 20 blocks; in every block both configurations were measured back to back (order alternating each block), each with 50 legacy + 50 hybrid connections per scenario, so both saw the same network conditions.
- **1000 measurements per configuration, mode and scenario; 8000 connections, 0 failures.** Every hybrid block was checked: X25519MLKEM768 negotiated in all 40 hackathon-server hybrid blocks, ML-DSA-44 verified in all 20 own-server hybrid blocks of each configuration.
- Idle temperature of both configurations measured back to back (120 s idle each) before the run; temperature recorded at the start of every block. The CPU setting was restored to the default automatically at the end.
- Files: `combined_raw.csv` (`config,scenario,mode,block,run,ms`), `combined_temps.csv`, `combined_stats.txt`, `combined.log`, `scripts/bench_combined.sh`.

Results

Own server (LAN), end-to-end time, full hybrid authentication

Config	Mode	Best	Median	Mean	p95	Worst
BASE	Legacy (X25519 + RSA)	12.64	**21.04**	21.17	24.19	35.08 ms
BASE	Hybrid (ML-KEM + ECDSA + ML-DSA-44, v1)	20.40	**28.06**	28.25	31.80	107.13 ms
COMBINED	Legacy (X25519 + RSA)	8.96	**15.35**	15.39	17.79	21.39 ms
COMBINED	**Hybrid (ML-KEM + ECDSA + ML-DSA-44, v2 two cores)**	**12.21**	**14.47**	**14.74**	**17.41**	**26.57 ms**

Hackathon server (IPv4), TLS handshake time

Config	Mode	Best	Median	Mean	p95	Worst
BASE	Legacy (X25519)	28.19	**38.36**	42.59	60.48	354.25 ms
BASE	Hybrid (X25519MLKEM768)	31.40	**43.76**	48.15	71.96	356.88 ms
COMBINED	Legacy (X25519)	25.90	**32.99**	36.76	55.41	355.01 ms
COMBINED	**Hybrid (X25519MLKEM768)**	**27.73**	**35.62**	**39.33**	**58.86**	**140.84 ms**

Target check: COMBINED hybrid vs the legacy BASELINE (paired per block)

Scenario	Baseline: legacy (BASE)	Hybrid (COMBINED)	Change	Hybrid faster in
Own server, full hybrid auth, end-to-end	21.04 ms	**14.47 ms**	**–6.57 ms (–31.2 %)**	**20/20 blocks**
Hackathon server, handshake	38.36 ms	**35.62 ms**	**–2.73 ms (–7.1 %)**	**17/20 blocks**

What COMBINED changes for each mode (paired per block)

Scenario	Mode	BASE → COMBINED	Faster in
Own server	Hybrid	28.06 → 14.47 ms (**–13.59 ms, –48.4 %**)	20/20
Own server	Legacy	21.04 → 15.35 ms (–5.70 ms, –27.1 %)	20/20
Hackathon server	Hybrid	43.76 → 35.62 ms (**–8.14 ms, –18.6 %**)	20/20
Hackathon server	Legacy	38.36 → 32.99 ms (–5.36 ms, –14.0 %)	18/20

Hybrid – legacy within the same configuration: BASE +7.15 ms (own) / +5.58 ms (hackathon); COMBINED **–0.99 ms (own, hybrid faster in 20/20 blocks)** / +2.84 ms (hackathon).

Temperature (°C)

Config	Idle after 120 s	During the run (start of each block) mean / max
BASE	45.9	47.8 / 48.9
COMBINED	47.2	47.8 / 48.5

For reference: full CPU speed (`performance`, fixed 1.5 GHz) idled at ~50–52 °C (2026-11-10).

How the approaches add up (own server, full hybrid, end-to-end median)

Configuration	Hybrid median	Source
Baseline (default clock, plain build, v1, one core)	27.95 ms (28.06 ms in this run)	`baseline_REPORT.md`, this run
Approach 2 only: bitmanip build	26.57 ms	`bitmanip_REPORT.md`
Approach 3 only: ML-DSA on a second core (v2)	24.57 ms	`multicore_e2e_REPORT.md`
Approach 1 only: min 750 MHz	16.19 ms	`dynamic_cpu_freq_REPORT.md`
All three combined	**14.47 ms**	this report
Legacy baseline, for comparison	*21.04–21.68 ms*	all runs

The single-approach rows come from different runs (each with its own alternating comparison), so compare them as an indication; the clean comparison is BASE vs COMBINED in this run. The approaches don't add up linearly: raising the minimum clock removes most of the slowdown, and the other two then shorten the remaining crypto work.

Findings

1. **The target is met with margin.** With all three approaches, full hybrid post-quantum TLS (ML-KEM-768 key exchange + ECDSA/ML-DSA-44 dual-signature authentication) is **31 % faster than the legacy baseline** on the own server (14.47 vs 21.04 ms, 20/20 blocks), and the hybrid key exchange against the hackathon server is **7 % faster than the legacy baseline** (35.62 vs 38.36 ms, 17/20 blocks).
2. **Under identical settings hybrid is no slower than legacy on the own server:** with COMBINED applied to both, hybrid is 0.99 ms faster than legacy (20/20 blocks), mostly because the server's RSA-3072 signing is slower than its ECDSA + ML-DSA-44 signing. Against the internet server the larger ML-KEM messages still cost +2.8 ms under identical settings.
3. **Hybrid gains twice as much as legacy from the same changes** (–13.6 vs –5.7 ms on the own server), because hybrid has more client-side crypto, which is what the three approaches speed up.
4. **Small heat cost:** +1.3 °C at idle (47.2 vs 45.9 °C); no difference during the run (both 47.8 °C mean). Full speed would cost ~+5 °C.
5. **Software and settings only:** no hardware accelerator, no new chip. The changes are a CPU clock setting, a compiler flag and a different place/core for the ML-DSA check.

Limits

- Hybrid authentication was measured against our own server; FOCAL's hackathon server supports the hybrid key exchange but has only an RSA-3072 certificate.
- The v2 ML-DSA signature uses a private-use TLS extension (`0xFE20`), not a standard; see `multicore_e2e_REPORT.md`.
- The bitmanip build runs only on CPUs with Zba/Zbb.
- The key exchange halves (X25519 ‖ ML-KEM) are still computed one after another inside OpenSSL (possible further gain of ~0.5–1 ms at 750 MHz, see `multicore_feasibility_REPORT.md`).
- Energy wasn't measured; temperature is the proxy.
- The 750 MHz minimum isn't made permanent on the board; it was active only during the measurements.

Source: combined_REPORT.md